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# Dynamic and wave propagation analysis of periodic smart beams coupled with resonant shunt circuits: passive property modulation

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**Abstract** The present study deals with analyzing the dynamic behavior of a smart beam coupled with piezoelectric materials in the unimorph and bimorph configurations connected to resonant shunt circuits. The resonant shunt circuit connection is applied to promote and passively modulate the vibration and flexural wave propagation attenuation. A Spectral Element Method (SEM)-based approach for the periodic smart structure is developed that is capable of efficiently analyzing various configurations of periodicity and piezoelectric attachments in a generic framework. The wave attenuation is obtained based on the Transfer Matrix Method (TMM). Numerical analyses reveal that the effective wavenumber presents local attenuation behavior at the same frequencies of the vibration as per the shunt design, indicating the shunt circuit's impedance associated with tuned frequency. The internal resonance characteristics of the resonant shunt circuits and their effects on the beam response are investigated in detail. The proposed periodic smart beam system along with the spectral element approach for analyzing the dynamics and wave propagation behavior of such systems, lays out further potential avenues for designing more complex configurations of smart structures and multifunctional metamaterials.

## 1 Introduction

The use of piezoelectric materials (piezo) coupled with structures to control vibration and wave propagation is increasingly applied in various advanced multi-physical systems. Such active control involves a direct piezoelectric effect, which modifies the properties of rigidity and damping of the structure by adding circuits with different combinations of resistive, inductive, capacitive and switch elements [1]. Therefore, it results in changes in the state variables of a physical, mechanical, electrical domain [2,3] and vice versa. These systems allow the evolution of smart materials, where on-demand property modulation potentiality can be obtained through passive, active and hybrid control [4]. However, for an electrical circuit composed of passive components (shunt circuit) connected to a piezo, there is a dissipation relationship of the electrical energy carried by the piezo conversion [5]. Such passive control shunt circuits can be used to control unwanted vibrations and noise [6]. Passive control techniques rely on changes in the basic structure without requiring a source of external electrical energy. In this case, the piezo sensors are connected with external electrical circuits composed of passive components, also known as shunt circuits [7].

Smart structures with resistive–inductive (RL) shunt circuit vibrational controls, also known as resonant shunt, imply similar effects to dynamic response as the mechanical vibration absorber [8]. Therefore, an electrical resonance can be tuned in different frequencies, thanks to the inductive element. Resonant shunt circuit's ability to attenuate the vibration of a structure at the tuning frequency is not limited by geometric, vibrational variations and dependence on external electrical energy [5,9]. Passive control uses external shunt circuits coupled to piezo, which can perform vibrational control without the external power supply. Forward [5] was the pioneer in associating some piezoelectric shunt circuits with the systems to induce vibrational control. Despite the wide variety of circuit topographies and their respective effects, the contribution of resistive and inductive shunts was discussed by Hagood and Flotow [8]. The purely resistive (R) shunts were related to structural damping bands, and RL circuits were related to controlling vibration modes, similar to a dynamic absorber. Shunt circuit configurations involving capacitive elements denote changes in the natural frequencies of the coupled system [10]. Wu [11] investigated the resonant RL shunt circuit by combining

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resistive and inductive components assembled in series and parallel circuit configurations. Hollkamp and Gordon [12,13] compared the efficiency in vibration control of resonant shunts to the viscoelastic materials and pointed the versatility of resonant shunts due to their possibility of adaptive tuning. Viana and Steffen [7] showed that the resonant RL shunt circuit is analogous to the one of a viscously damped dynamic vibration absorber. Periodicity in the smart structure has the potential to induce bandgap effects on vibrating structures. It includes techniques such as multimodal [9], negative capacitance [14], and variations in the resistive–inductive–capacitive (RLC) circuit [15]. A comprehensive review of shunt piezoelectric systems applied in noise and vibration control was published in [16] .The use of smart material gained renewed interest in applications involving vibration-based energy harvesting, non-linearities, and in the framework of metamaterials [17,18]. Such smart metamaterials can result in adaptive and multifunctional materials.

Piezo shunt circuits in vibrations control have been used in turbine blades [19], CFM56-7B turbojet blade [20], Boeing's helicopter blade [21], car's chassis [22], centrifuge [23], smart gloves [24], and various other structural elements [9,25]. Numerical methods applied to investigate vibration and wave using smart structure goes from modal analysis framework [26,27] and experimental tests [28] to state-space formulation [29], and the finite element method (FEM) [25,30]. Most of these require many elements in the mesh, as FEM, which makes the computational analysis intensive and time-consuming. An alternative in dynamic simulation and analysis is the spectral element method (SEM). This method is formulated from the wave equation's analytical solution reducing the number of elements in the simulation significantly. Therefore, a single element is sufficient to model a smart material with a uniform section along its length. The element is tailored with the matrix ideas of the finite element method (FEM), where the interpolation function is the exact solution of wave equation A few spectral elements have been developed, like a rod, beam, plates, [31–33], cables [34–36], and ongoing researches are proposing new and improved elements. Recently the spectral elements of a beam-like element have been exploited to develop closed-form solutions for the dynamic behavior of lattice metamaterials, leading to the characterization of frequency-dependent effective elastic moduli [37,38]. Lee and Kim [39,40] developed a spectral element coupling the Euler–Bernoulli beam with a piezoelectric layer components and in [41] with two piezoelectric layers. A smart unimorph beam spectral element with passive control was proposed and demonstrated in [42]. In this direction, there is a lack of numerical models to formulate the spectral elements for bimorph beams coupled to the shunt circuits. Despite numerous applications and numerical models exploring vibration control based on the shunt techniques, a formulation regarding the spectral approach for bimorph beam connected to a shunt circuit passive control is not existent in the literature. To date, only a small number of studies have demonstrated elastic wave attenuation derived from this type of system control.

The present work explores functional materials under passive property modulation, which can attenuate vibration and propagate elastic waves. Smart structure design and the dynamic solution obtained using low computational cost with high accuracy are the key issues here. Therefore, this paper aims to investigate the elastic wave propagation and the beam's dynamic characteristic under the shunt circuit control. The novel contributions of this article are: (1) to investigate the physical state of elastic wave propagation and disruption due to the electrical resonance input by the shunt circuit; (2) to explore the effects of RL, LC, and RCL shunts for controlling structural vibration response along with wave propagation and attenuation; (3) to propose a formulation of bimorph beam spectral elements coupled with shunt circuits. The smart beam's dynamic behavior is coupled with a layer of piezo (unimorph), or two layers of piezo (bimorph), both connected to the resonant shunt circuits in the configuration of inductive–capacitive (LC), RL, and RLC. Here the numerical models would be developed based on SEM to carry out an accurate yet computationally efficient analysis without fine discretization. The transfer matrix method would be used to estimate dispersion diagrams and the attenuation properties of unimorph and bimorph beams.

The paper is organized hereafter as follows. The unimorph and bimorph smart spectral elements are introduced in Sect. 2. In Sect. 3, the formulation of resonant shunt circuits (to promote the beam's control) is presented, and in Annex 1 the transfer matrix method (used to estimate the dispersion curves) is explained. The dynamic properties of the smart beams are investigated numerically in Sect. 4, and the conclusion is drawn in Sect. 5.

## 2 Piezo-beam spectral element

### 2.1 Unimorph beam spectral element

One of the major advantages of Spectral Element Method (SEM) is the small number of elements needed to represent a system. This feature leads to a significant computational efficiency level in dynamic and wave propagation analyses where a fine discretization is normally required following the conventional finite element-based approaches. The element shape functions are obtained from the analytical solution of governing differential equations and the dynamic system solutions are written in the frequency domain. This characteristic improves the accuracy of the dynamic system solution. In addition, a wave-like interpretation of the structural dynamics can be straightforwardly obtained, leading to innovative wave-based control strategies. Figure 1 shows the geometry and spectral element node reactions that are considered in analyzing the smart beam with unimorph configuration.

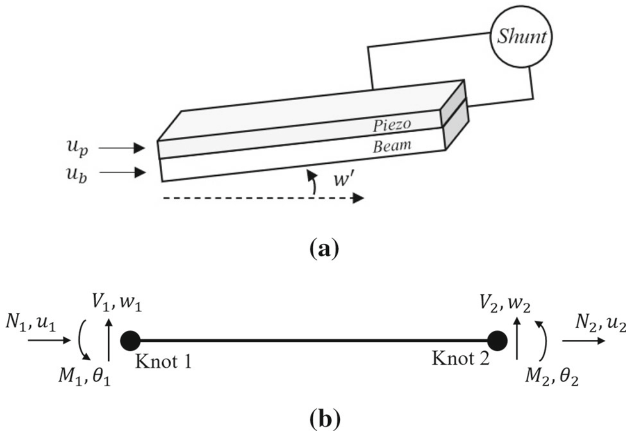



**Fig. 1** Smart unimorph beam: **a** physical structure; **b** spectral element model

SEM can represent the smart beam with only two nodes, as shown in Fig. 1b. Each node has its nodal forces and displacements following the Euler–Bernoulli theory to represent the beam that is equivalent to a piezoelectric transducer [32]. The thickness and displacement relationships of the beam with piezoelectric material are defined as

$$u_{\mathrm{p}} = u_{\mathrm{b}} - \frac{h}{2}\frac{\partial w}{\partial x}, \tag{1}$$

where $u_{\mathrm{p}}$ and $u_{\mathrm{b}}$ are the axial displacements of the piezo and the beam, respectively. The transverse displacement is $w$, and $h = h_p + h_b$ is the sum of thickness of the piezo and the beam. By using the constitutive relation the strain energy of the two-layer beam is derived after integrating over its base; the potential and kinetic energy applied into Hamilton's principle can be described as

$$\int_{t_1}^{t_2} [\delta(T - U) + \delta W] = 0, \tag{2}$$

where the potential energy is

$$U = \frac{1}{2}\int_0^L \left(E_b A_b u_b'^2 + E_b I_b w''^2 + C_{11}^D A_p u_p'^2 \right. \\ \left. + C_{11}^D I_p w''^2 - 2A_p h_{31} D u_p' + A_p \beta_{33}^S D^2\right) \mathrm{d}x \tag{3}$$

and the kinetic energy is

$$T = \frac{1}{2}\int_0^L \left\{\rho_b A_b \left(\dot{u}_b^2 + \dot{w}^2\right) + \rho_p A_p \left(\dot{u}_p^2 + \dot{w}^2\right)\right\} \mathrm{d}x, \tag{4}$$

where $E$, $A$, $I$ and $\rho$ (for each layer) are Young's modulus, the cross-sectional area, the area moment of inertia about the neutral axis, and the mass density, respectively. The virtual work due to the non-conservative forces is given by

$$\delta W = \int_0^L b\mathtt{V}(t)\, \delta D dx + N \delta u_b \mid_L^0 \\ + M \delta\theta \mid_L^0 + Q \delta w \mid_L^0, \tag{5}$$

where ($\prime$) denotes space derivative, ($\dot{}$) denotes time derivative, and damping is presented by $c$. By applying Eqs. (3) and (4) into the Hamilton's principle, the axial-bending coupled equation of motion [39] is given as follows

$$EIw'''' + \rho A\ddot{w} + cA\ddot{w} = -\alpha\ddot{u}_b' + \beta u''' + \lambda\ddot{w}'' \\ + c_1\ddot{w}'' - c_4\ddot{u}_b' + Fw'' + p(x,t), \\ EAu_b'' - \rho A\ddot{u}_b - cA\ddot{u}_b = -\alpha\ddot{w}' + \beta w''' - \tau(x,t), \tag{6}$$

where

$$EA = E_b A_b + E_p A_p, \qquad EI = E_b I_b + E_p I_p + (1/4)\, E_p A_p h^2, \\ c_4 = (1/2) c_p A_p h, \\ \rho A = \rho_b A_b + \rho_p A_p, \qquad E_p = C_{11}^D - h_{31}^2 \beta_{33}^{S-1}, \\ cA = c_b A_b + c_p A_p, \\ \gamma = (1/4)\rho_p A_p h^2, \quad \alpha = (1/2)\rho_p A_p h, \\ \beta = (1/2) E_p A_p h, \quad c_1 = (1/4) c_p A_p h^2.$$

Since SEM is formulated in the frequency domain, the Fourier transform is applied to convert the governing equations in the frequency domain. Hence, the spectral element assumption is that displacements and voltage $V(t)$ [43] are expressed spectrally as

$$w(x,t) = \sum_n^N \hat{W}(x,\omega_n) e^{-i\omega_n t}, \tag{7a}$$

$$u(x,t) = \sum_n^N \hat{U}(x,\omega_n) e^{-i\omega_n t}, \tag{7b}$$

$$V(t) = \sum_n^N \hat{V}(\omega_n) e^{-i\omega n t}. \tag{7c}$$

By replacing the spectral components of Eq. (7) in the equation of motion of the smart beam i.e. Eq. (6), the electromechanical equation of motion in the frequency domain is given as

$$EI\hat{W}''' - \omega^2\rho A\hat{W} = \omega^2(-\gamma\hat{W}'' + \alpha\hat{U}') + \beta\hat{U}''', \\ EA\hat{U}'' + \omega^2\rho A\hat{U} = \omega^2\alpha\hat{W}' + \beta\hat{W}''''. \tag{8}$$

The general solution of the spectral displacements follows the relationship

$$\hat{W}(x) = \sum_{i=1}^{6} a_j e^{-ik_j x} = \mathbf{e}(x,\omega)\mathbf{a}, \tag{9a}$$

$$\hat{U}(x) = \sum_{i=1}^{6} r_j a_j e^{-ik_j x} = \mathbf{e}(x,\omega) R\mathbf{a}, \tag{9b}$$

where

$$\mathbf{e}(x,\omega) = [e^{-ik_1 x}, e^{-ik_2 x}, e^{-ik_3 x}, e^{-ik_4 x}, e^{-ik_5 x}, e^{-ik_6 x}],$$
$$\mathbf{a} = [a_1, a_2, a_3, a_4, a_5, a_6]^T$$
$$R = \mathrm{diag}\left(\frac{-\omega k_j c_4 - i\omega^2 k_j \alpha + ik_j^3 \beta}{-k_j^2 EA + \omega^2 \rho A - i\omega c A}\right).$$

A characteristic equation with an eigenvalue problem is obtained from the general solution of the smart material structure equation of motion, where the wave numbers $k_i (i = 1, 2, 3 \ldots, 6)$ can be determined, as expressed in [32]. By considering a finite structure with length $L_e$, the spectral nodal displacements in terms of $a_j$, which satisfy the boundary conditions of the system, can be represented by the vector $x$ as

$$x = \mathbf{H}(\omega) a = \left[\hat{U}_1\ \hat{W}_1\ \hat{\theta}_1\ \hat{U}_2\ \hat{W}_2\ \hat{\theta}_2\right]^T, \tag{10}$$

where

$$\mathbf{H}(\omega) = \begin{bmatrix} r_1 & r_2 & r_3 & r_4 & r_5 & r_6 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ -ik_1 & -ik_2 & -ik_3 & -ik_4 & -ik_5 & -ik_6 \\ e^{-ik_1 L_e} r_1 & e^{-ik_2 L_e} r_2 & e^{-ik_3 L_e} r_3 & e^{-ik_4 L_e} r_4 & e^{-ik_5 L_e} r_5 & e^{-ik_6 L_e} r_6 \\ e^{-ik_1 L_e} r_1 & e^{-ik_2 L_e} & e^{-ik_3 L_e} & e^{-ik_4 e L} & e^{-ik_5 L_e} & e^{-ik_6 L_e} \\ -ik_1 e^{-ik_1 L_e} & -ik_2 e^{-ik_2 L_e} & -ik_3 e^{-ik_3 L_e} & -ik_4 e^{-ik_4 L_e} & -ik_5 e^{-ik_5 L_e} & -ik_6 e^{-ik_6 L_e} \end{bmatrix}.$$

The dynamic interaction of each nodal component leads to the following dynamic stiffness matrix of the unimorph spectral beam element

$$\mathbf{S}(\omega) = \mathbf{H}^{-1}(\omega)\mathbf{D}(\omega)\mathbf{H}^{-1}(\omega), \tag{11}$$

where

$$\begin{aligned}\mathbf{D}(\omega) = &-\mathrm{EARKEKR} + \mathrm{EIK^2EK^2} \\ &- i\beta(\mathrm{K^2EKR} + \mathrm{RKEK^2}) \\ &- \omega^2[\rho A(\mathrm{E} + \mathrm{RER}) + i\alpha(\mathrm{KER} + \mathrm{REK}) - \lambda \mathrm{KEK}] \\ &+ i\omega[cA(\mathrm{E} + \mathrm{RER}) - c_1 \mathrm{KEK} \\ &+ ic_4(\mathrm{KER} + \mathrm{REK})] - \mathrm{FRER}\end{aligned}$$
$$K^2 = (K = \mathrm{diag}(k_i))^2,$$
$$E(\omega) = \int_0^L \mathbf{e}^T(x,\omega)\mathbf{e}(x,\omega)\mathrm{d}x.$$

The dynamic relationship of the piezoelectric beam can be expressed by $\mathbf{S}(\omega)$, which is the frequency-dependent dynamic stiffness matrix.

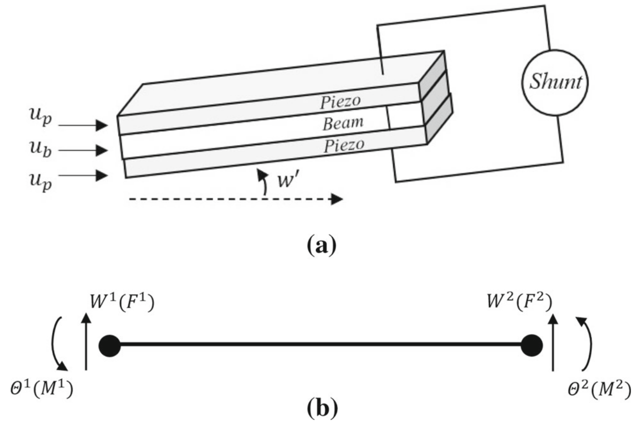



**Fig. 2** Smart bimorph beam: **a** physical structure; **b** spectral element model

## 2.2 Bimorph beam spectral element

The bimorph beam showed in Fig. 2a is composed by a beam with two piezo layer follows the constitutive equation expressed in Eq. (30).

The potential energy for the bimorph beam is

$$U = \frac{1}{2}\int_0^L \left[\kappa GA(\phi - W'')D_t\phi'' + \kappa W' - \kappa GA\Phi + V - g(\Theta(L_c) - \Theta(0))\right]\mathrm{d}x \tag{12}$$

and the kinetic energy is

$$T = \frac{1}{2}\int_0^L \left[\rho I\omega^2\Theta - \rho I\omega^2 W\right]\mathrm{d}x. \tag{13}$$

By applying Eqs. (12) and (13) into the Hamilton's principle, the equation of motion of a bimorph beam in a frequency domain can be expressed as [41]

$$\begin{aligned}&\kappa \mathrm{GA}\left[\frac{\partial\Theta}{\partial x} - \frac{\partial^2 W}{\partial x^2}\right] - \rho A\omega^2 W = 0, \\ &D_t\frac{\partial^2\Theta}{\partial x^2} + \kappa\mathrm{GA}\frac{\partial W}{\partial x} + (\rho I\omega^2 - \kappa\mathrm{GA})\Theta = 0, \\ &V = g(\Theta(L_e) - \Theta(0)),\end{aligned} \tag{14}$$

where $g$ is the electrical relation expressed by

$$g = \frac{i\omega R d_{31}\bar{h}b}{S_{11}^E(1+i\omega RC_p)} \tag{15}$$

and

$$\begin{aligned}
&\mathrm{GA} = G_b b h_b + 2G_p b h_p, \rho A = \rho_b b h_b + 2\rho_p b h_p,\\
&\rho I = \rho_b I_b + 2\rho_p I_p,\\
&D_t = D_{t1} = E_b I_b + 2C_{11}^D I_p - 2A_p \bar{h}^2 h_{31}^2 \beta_{33},\\
\text{or} &D_t = D_{t2} = E_b I_b + 2C_{11}^E I_p.
\end{aligned}$$

Here, $\kappa$ is the shear factor, $G$ is the shear modulus, $D_{t1}$ and $D_{t2}$ are the electrical components in open and short circuit, respectively. The general solution of the spectral displacements follows the relationship

$$W(x) = \sum_{j=1}^{2} a_j e^{-ik_j x} = \mathbf{g}(x,\omega)\mathbf{a}, \tag{16a}$$

$$\Theta(x) = \sum_{j=1}^{4} r_j a_j e^{ik_j x} = \mathbf{g}(x,\omega) r_i \mathbf{a}, \tag{16b}$$

where $a_j$ and $r_j$ are wave modes coefficients, and $k_j$ is the wavenumber defined as

$$\pm\, k_t = \sqrt{\frac{\kappa GA\rho I\omega^2 + D_t\rho A\omega^2 + \sqrt{(\kappa \mathrm{GA}\rho I\omega^2 + D_t\rho A\omega^2)^2 - 4D_t\kappa\mathrm{GA}\rho A\omega^2(\rho I\omega^2 - \kappa\mathrm{GA})}}{2D_t\kappa\mathrm{GA}}}, \tag{17}$$

$$\pm k_e = \sqrt{\frac{\kappa \mathrm{GA}\rho I\omega^2 + D_t\rho A\omega^2 - \sqrt{(\kappa \mathrm{GA}\rho I\omega^2 + D_t\rho A\omega^2)^2 - 4D_t\kappa\mathrm{GA}\rho A\omega^2(\rho I\omega^2 - \kappa\mathrm{GA})}}{2D_t\kappa\mathrm{GA}}}. \tag{18}$$

Based on the solutions of the wavenumber, the displacement and rotation have the following forms

$$W(x) = a_1 e^{ik_1 x} + a_2 e^{ik_2 x} + a_3 e^{ik_3 x} + a_4 e^{ik_4 x}, \tag{19}$$

$$\begin{aligned}\Theta(x) = {} & R_1 a_1 e^{ik_1 x} + R_2 a_2 e^{ik_2 x}\\ & + R_3 a_3 e^{ik_3 x} + R_4 a_4 e^{ik_4 x}.\end{aligned} \tag{20}$$

From the characteristic equations, one can also obtain the coefficient $r$ as

$$\begin{aligned}
r_j &= ik_j + \frac{\rho A\omega^2}{i\kappa GAk_j}, \qquad (j = 1,2,3,4.)\\
\mathbf{R} &= \mathrm{diag}(r_j),\\
\mathbf{e}_W(x,\omega) &= [e^{ik_1 x}, e^{ik_2 x}, e^{ik_3 x}, e^{ik_4 x}],\\
\mathbf{e}_\Theta(x,\omega) &= \mathbf{R}\mathbf{e}_W(x,\omega), \quad \mathbf{a} = [a_1, a_2, a_3, a_4]^T.
\end{aligned} \tag{21}$$

The spectral nodal displacements and slopes of the bimorph beam element can be related to the displacement field by

$$\mathbf{d} = \begin{bmatrix} W_1(x=0) \\ \Theta_1(x=0) \\ W_2(x=L) \\ \Theta_2(x=L) \end{bmatrix} = \begin{bmatrix} \mathbf{e}_W(0,\omega) \\ \mathbf{e}_\Theta(0,\omega) \\ \mathbf{e}_W(L,\omega) \\ \mathbf{e}_\Theta(L,\omega) \end{bmatrix} \mathbf{a} = \mathbf{H}_{pvp}(\omega)\mathbf{a}, \tag{22}$$

where by applying the boundary condition at the nodes one has

$$\mathbf{H}_{pvp}(\omega) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ r_1 & r_2 & r_3 & r_4 \\ e^{ik_1 L_e} & e^{ik_2 L_e} & e^{ik_3 L_e} & e^{ik_4 L_e} \\ r_1 e^{ik_1 L_e} & r_2 e^{ik_2 L_e} & r_3 e^{ik_3 L_e} & r_4 e^{ik_4 L_e} \end{bmatrix}.$$

The spectral components of the bending moment and transverse shear at the nodes are

$$\begin{aligned}
F &= \kappa\mathrm{GA}\left(\frac{\partial W}{\partial x} - \Theta\right),\\
M &= D_t \frac{\partial \Theta}{\partial x} + \frac{\bar{h}bd_{31}}{S_{11}^E} V.
\end{aligned} \tag{23}$$

The spectral nodal forces can be related to the corresponding forces and moments defined at boundary condition as

$$\mathbf{f} = \begin{bmatrix} F_1 \\ M_1 \\ F_2 \\ M_2 \end{bmatrix} = \begin{bmatrix} F_1(x=0) \\ M_1(x=0) \\ F_2(x=L) \\ M_2(x=L) \end{bmatrix} = \mathbf{G}_{pvp}(\omega)\mathbf{a}, \tag{24}$$

where $\mathbf{G}_{pvp}(\omega)$, and $\mathbf{I}$ are given as

$$\mathbf{G}_{pvp}(\omega)=\begin{bmatrix} -\kappa GA(r_1-ik_1) & -\kappa GA(r_2-ik_2) & -\kappa GA(r_3-ik_3) & -\kappa GA(r_4-ik_4) \\ -D_t r_1 ik_1 & -D_t r_2 ik_2 & -D_t r_3 ik_3 & -D_t r_4 ik_4 \\ \kappa GA(r_1-ik_1)e^{ik_1L_e} & \kappa GA(r_2-ik_2)e^{ik_2L_e} & \kappa GA(r_3-ik_3)e^{ik_3L_e} & \kappa GA(r_j-ik_4)e^{ik_4L_e} \\ D_t r_1 ik_1 e^{ik_1L_e} & D_t r_2 ik_2 e^{ik_2L_e} & D_t r_3 ik_3 e^{ik_3L_e} & D_t r_4 ik_j e^{ik_4L_e} \end{bmatrix},$$

$$\mathbf{I}=\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}.$$

The relation between spectral nodal displacements and forces is given by the coefficients $a_i$, that are defined as the frequency-dependent dynamic stiffness matrix. It relates the nodal spectral forces with the nodal spectral displacements expressed as $\mathbf{f}=\mathbf{S}(\omega)\mathbf{d}$, where

$$\mathbf{S}(\omega)=\mathbf{G}_{pvp}(\omega)\mathbf{H}_{pvp}^{-1}(\omega)+g\frac{\bar{h}bd_{31}}{S_{11}^{E}}\mathbf{I}, \tag{25}$$

where $\mathbf{S}(\omega)$ is the frequency-dependent bimorph beam spectral element matrix or dynamic stiffness matrix given in [41].

### 2.3 Unimorph and bimorph beams coupling with shunt circuit

The global electromechanical equation of motion coupling the shunt circuit to the piezoelectric is defined in terms of the spectral stiffness matrix as [42],

$$\begin{aligned} &\mathbf{S}(\omega)x-\mathbf{S}_{\mathrm{SH}}(\omega)V(\omega)=f(\omega), \\ &i\omega\mathbf{S}_{\mathrm{SH}}(\omega)x+i\omega C_p^T V(\omega)=I(\omega), \end{aligned} \tag{26}$$

where $\mathbf{S}_{\mathrm{SH}}$ is the shunt circuit spectral matrix, $x$ is the generalized nodal displacement, $f$ the generalized force, $I$ is the current in the spectral domain, and $V$ is the voltage. Similar to the nodal relation in Eq. (10), the behavior of the piezoelectric structure with the branch circuit can be expressed with the displacements and equivalent nodal forces. Therefore, $\mathbf{S}_{\mathrm{SH}}$ can be assembled as follows

$$\begin{aligned}\mathbf{S}_{\mathrm{SH}}(\omega)=\big[&N_{e1}W(x_0,\omega),0,-M_{e1}W(x_0,\omega), \\ &-N_{e2}W(x_0,\omega),0,M_{e2}W(x_0,\omega)\big]^T,\end{aligned} \tag{27}$$

where

$$\begin{aligned} N_{e1}=N_{e2}&=\frac{k_{ij}^2 i\omega Z^{EL} b_p d_{31}E_p}{1+i\omega C_p^T Z^{EL}}, \\ M_{e1}=M_{e2}&=\frac{k_{ij}^2 i\omega Z^{EL} h b_p d_{31}E_p}{2+2i\omega C_p^T Z^{EL}}. \end{aligned}$$

The nodal functions of the piezoelectric structure with shunt circuit are related to piezoelectric coupling coefficient $k_{ij}$, the width $b_p$, Young's modulus $E_p$, and the piezoelectric constant $d_{31}$. Therefore, a general representation of the dynamic behavior of the smart beam can be given as

$$\left[\mathbf{S}(\omega)+\omega^2\mathbf{S}_{SH}^2\left(\tfrac{1}{i\omega+1/Z^{EL}}\right)\right]x(\omega)=f(\omega) \tag{28}$$

and $Z^{\mathrm{EL}}$ is the associated impedance to each shunt circuit configuration. Section 3 presents an overview of the resonant shunt circuit impedances in different configurations, as explored in this paper to modulate the vibration control and wave propagation.

## 3 Impedances of resonate shunt circuits

The electro-mechanical energy conversion or vice versa follows the constitutive equation [8,44]; for the unimorph configuration it is given as

$$\begin{bmatrix} D_3 \\ \epsilon_p \end{bmatrix}=\begin{bmatrix} \epsilon_{33}^T & d_{31} \\ d_{31} & S_{11}^E \end{bmatrix}\begin{bmatrix} E_3 \\ \sigma_p \end{bmatrix}. \tag{29}$$

The electrical circuit can be connected in a series or parallel configuration with the upper and lower piezoelectric layers, as shown in Fig. 3b and c, respectively. The assumption of each configuration impact the structure's rigidity because of different interactions between capacitances and piezoelectric $C_p^T$ [41]. Hence, the constitutive equation for the bimorph configuration is

$$\begin{bmatrix} \epsilon_p \\ \gamma_p \\ D_3 \end{bmatrix}=\begin{bmatrix} S_{11}^E & 0 & d_{31} \\ 0 & S_{55}^E & 0 \\ d_{31} & 0 & \epsilon_{33}^T \end{bmatrix}\begin{bmatrix} \sigma_p \\ \tau_p \\ E_3 \end{bmatrix}, \tag{30}$$

where $\epsilon_p$ is the normal strain, $\gamma_p$ is the shear strain, $D_3=Q/A$ [C/m$^2$] is the electric displacement. The quantity $Q$ represents the electric change accumulated in the electrodes of the piezo surface, $\sigma_p$ is the normal stress, $\tau_p$ the shear stress, $E_3=V/l$ [V/m] is the electric field, $\epsilon_{33}^T$ is the permittivity constant, $d_{31}$ [m/V] is piezoelectric constant, $S_{11}^E$ and $S_{55}^E$ [m$^2$/N] are the mechanical elastic compliance constants. Subscripts $\bullet^T$ and $\bullet^E$ indicate mechanical stress and electrical field, respectively. By applying Laplace transformation in Eq. (29) and assuming voltage-current relation,

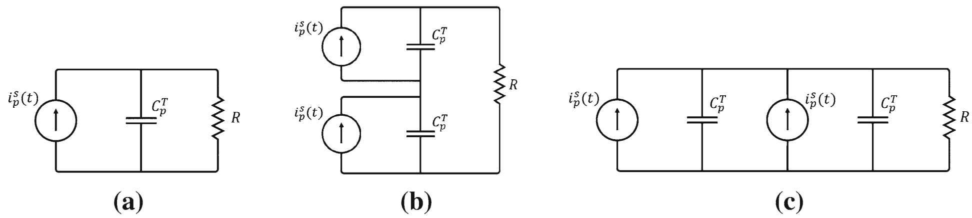



**Fig. 3** Topographic circuit representation of piezoelectrics connected with a resistor in the configuration: **a** Unimorph beam; **b** Bimorph beam in series; **c** Bimorph beam in parallel

one obtains the piezo expression

$$\begin{bmatrix} I \\ \epsilon_p \end{bmatrix} \begin{bmatrix} Y^{\mathrm{EL}} & s\mathcal{A}d \\ d_{31}L^{-1} & S_{11}^E \end{bmatrix} = \begin{bmatrix} V \\ \sigma_p \end{bmatrix}, \tag{31}$$

where $Y^{\mathrm{EL}}$ is the electrical admittance, $L^{-1}$ is a diagonal matrix with the piezo length, $\mathcal{A}$ is a diagonal matrix with surface area and $s$ the Laplace parameter. The general electrical admittance is obtained by the sum of the piezo's admittance, $Y^P$, with the external shunt circuit $Y^{\mathrm{SH}}$, as follows

$$Y^{\mathrm{EL}} = Y^P + Y^{\mathrm{SH}}. \tag{32}$$

The equivalent admittance of the piezo is $Y^{\mathrm{EL}} = 1/Z^{\mathrm{EL}}$, which can lead to the relation [44] $s_{jj}^{SH} = s_{jj}^E[1 - k_{ij}^2 \bar{Z}_i^{\mathrm{EL}}]$. Here, $s_{jj}^{SH}$ is the shunt circuit compliance coupled with the piezo; $k_{ij} = d_{ij}(s_{jj}^E \in_i^T)^{1/2}$ is the electromechanical coupling coefficient, and $\bar{Z}_i^{\mathrm{EL}}$ is he impedance matrix. These coupling relations and the impedance influence Young's modulus of the structure, as shown in the following expression

$$E_{jj}^{\mathrm{SH}} = E_{jj}^D \frac{1 - k_{ij}^2}{1 - k_{ij}^2 \bar{Z}_i^{\mathrm{EL}}}, \tag{33}$$

where $E_{jj}^D$ and $E_{jj}^{\mathrm{SH}}$ are the Young's modulus of the piezo and the shunt circuit, respectively. Hence, the structure piezo coupling influence the dynamical characteristic [5].

Figure 3a–c shows the topographic circuit representation of piezo connected to a resistor in series and parallel configuration. The piezoelectric component has an internal capacitance effect inherent to its electromechanical property. The piezo capacitance assumes different values in each configuration, so that [41],

$$\text{Series:}\ C_p^T = \frac{\epsilon_{33}^S L_e b}{2h_p}, \tag{34a}$$

$$\text{Parallel:}\ C_p^T = 2\frac{\epsilon_{33}^S L_e b}{h_p}, \tag{34b}$$

where $b$ and $L_e$ are of the unimorph and bimorph beam's width and length, respectively; $h_p$ is piezo layer thickness, while the piezo compliance is given by $\epsilon_{33}^S = \epsilon_{33}^T - d_{31}^2/s_{11}^E$. The capacitance ($C_p^T$) between the piezoelectric surfaces perpendicular to the direction $i$ (in constant stress), the impedance of a shunt circuit depend on the arrangement of the components used in the circuit. In a passive control usage of the piezoelectric component, its capacitance is influenced by an open circuit, a short circuit, or an externally connected shunt circuit [8]. For the cases with constant stress, the short circuit and open circuit impedances (used in Eq. 27) will be

$$\text{Short:}\ Z^{EL} = 0, \tag{35a}$$

$$\text{Open:}\ Z^{EL} = i\omega C_p^T. \tag{35b}$$

In general, shunts circuits can be classified by the predominance of resistive, inductive, capacitive and switching elements. Figure 4 shows some of the main shunt circuits' topographies used to promote the effects of energy dissipation associated with vibration, noise and wave propagation control.

Components inside the dashed line in Fig. 4 is the piezoelectric representation, and outside are the shunt circuit elements. Figure 4a–c shows pure resistive with piezo admittance $Y^D = i\omega C_p^T$, inductive with the admittance ratio of $Y^{\mathrm{SH}} = 1/i\omega L$ where the real part equals to zero, and capacitive shunt circuit. Generalization of the electrical impedance of the resistive, inductive, and capacitive shunt circuit is given by

$$Z^{\mathrm{EL}} = \frac{1}{i\omega C_p^T + \frac{1}{R}}, \tag{36a}$$

$$Z^{\mathrm{EL}} = \frac{i\omega L}{-\omega^2 L C_p^T + 1}, \tag{36b}$$

$$Z^{\mathrm{EL}} = \frac{1}{i\omega C_p^T}. \tag{36c}$$

Capacitive shunt circuits, Fig. 4c, are commonly used in combination with other components to provide more efficiency in control of the total system stiffness [45]. However, applications with negative capacitive elements can promote attenuation and generate a bandgap effect in periodic structures [30]. For circuits with inductive and capacitive components, a predominant imaginary complex is established as a pass-through filter. In the circuit of Fig. 4d, a control in the frequency range is associated with admittance in series as $Y^{\mathrm{SH}} = 1/(i\omega L + 1/i\omega C)$ and parallel as $Y^{\mathrm{SH}} = 1/i\omega L + i\omega C$. Therefore, the general impedances

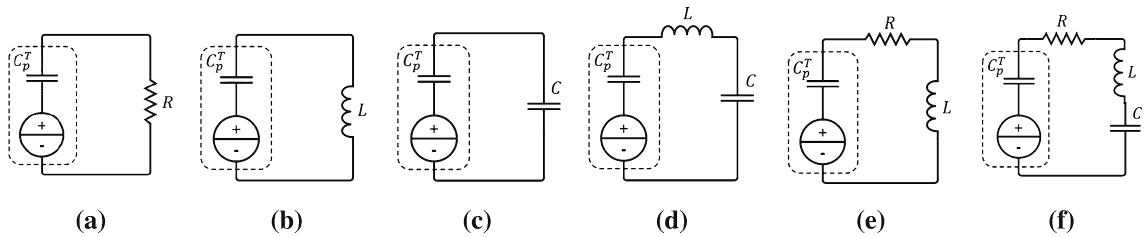



**Fig. 4** Shunt circuit topographies **a** pure resistive; **b** pure inductive; **c** pure capacitive; **d** LC; **e**) RL; **f** RLC

for the series and parallel circuit are given as

$$\text{Series: } Z^{\mathrm{EL}} = \frac{i\omega L + \frac{1}{i\omega C}}{-\omega^2 L C_p^T + \frac{i\omega C_p^T}{i\omega C} + 1}, \tag{37a}$$

$$\text{Parallel: } Z^{\mathrm{EL}} = \frac{i\omega L}{1 - \omega^2 LC - \omega^2 L C_p^T}. \tag{37b}$$

Resonant circuits always has incorporated an inductive component in its electronic part. The RL circuit of Fig. 4e is one of the most used shunts to promote passive vibration control because it works similar to a mechanical absorber without damping. Forward [5] investigated the inductive element as an absorber and observed an equivalent behavior when adding extra mass to the system. However, the inductance is inversely proportional to the frequency of the oscillations to be attenuated. The general impedances associated with the admittances of the series $Y^{\mathrm{SH}} = 1/(R + i\omega L)$ and parallel $Y^{\mathrm{SH}} = 1/i\omega L + 1/R$ circuit are represented as follows

$$\text{Series: } Z^{\mathrm{EL}} = \frac{R + i\omega L}{1 - \omega^2 L C_p^T + i\omega R C_p^T}, \tag{38a}$$

$$\text{Parallel: } Z^{\mathrm{EL}} = \frac{i\omega R L}{i\omega L + R - \omega^2 R L C_p^T}. \tag{38b}$$

Similar to RL and LC circuits, the RLC shunt circuit(refer to Fig. 4f) is a complete version of electrical components with specific properties for attenuation and vibration control. The admittances for the series $Y^{\mathrm{SH}} = 1/(R + i\omega L + 1/C)$ and parallel $Y^{\mathrm{SH}} = 1/R + 1/i\omega L + i\omega C$ circuit form the following impedances

$$\text{Series: } Z^{\mathrm{EL}} = \frac{R + i\omega L + \frac{1}{i\omega C}}{1 + i\omega C_p^T (R + i\omega L + \frac{1}{i\omega C})}, \tag{39a}$$

$$\text{Parallel: } Z^{\mathrm{EL}} = \frac{i\omega L}{i\omega L(i\omega C_p^T + i\omega C) + \frac{i\omega L}{R} + 1}. \tag{39b}$$

## 4 Results and discussion

Vibration attenuation performed with shunt circuit entails the main concept of transforming the dynamic strain energy (mechanical energy) of the host structure into electric energy and vice-versa. This technique connects electric or electronic circuits (represented by the impedance) to the electrodes of a piezoelectric material (usually the piezo transducer) bonded to a vibrating structure. The piezo guides the energy into the shunt circuit, where it is partially dissipated. The resonant shunt circuits are well known because of their similarities to the viscously damped dynamic vibration absorber. The circuit design represented by the impedance is chosen to minimize the structural vibration of a specific pre-identified mode shape.

Aside from the vibration, the elastic waves propagating in the media manifest disturbance or variation that transfers energy progressively from point to point in a solid structure, resulting in elastic deformation. The dispersion curves or diagram could display the propagation velocities of all possible guided wave modes at different frequencies in a structure. Dispersion curves can predict the change in frequency of the wave signals due to the inclusion of circuit shunt inducing some disturbance in the waveguide. Complex dispersion curves have their real part of wavenumber related to the positive-going flexural waves, and the imaginary part related to the non-propagating wave constant. The dispersion curves indicate where the disturbance is occurring and estimate the band attenuation.

This paper explores vibration and wave propagation phenomena under the control of three configurations of resonant shunt circuit. The analyses are performed in unimorph and bimorph smart beams coupled to the resonant shunt circuits in series and parallel setting. The circuit's components induce a great influence in the electrical impedance, which behaves like a mechanical system with mass-stiffness-damping. The resistor $R$ impacts on the system damping, the capacitor influences on the elastic properties, and the inductor $L$ is directly associated with mass that controls the tuning resonant frequency. Figure 5a–c illustrates the structures and unit-cell of the unimorph and bimorph beam-shunt in series and parallel configurations.

In the numerical analysis, we assume unimorph and bimorph periodic beams with a length of $L_e = 261.6$ mm, as shown in Fig. 5. Subscripts $\bullet_p$ and $\bullet_b$ denote beam and piezo, respectively. The width of both components are equal to $b_b = b_p = 12.7$ mm, and the thicknesses are $h_b = 2.286$ mm and $h_p = 0.762$ mm. Young's moduli are $E_b = 71$ GPa and $E_p = 64.9$ GPa, Shear moduli are $G_b = 23$ GPa and $G_p = 40$ GPa, densities are $\rho_b = 2700$ kg/m$^3$ and $\rho_p = 7600$ kg/m$^3$. Piezoelec-

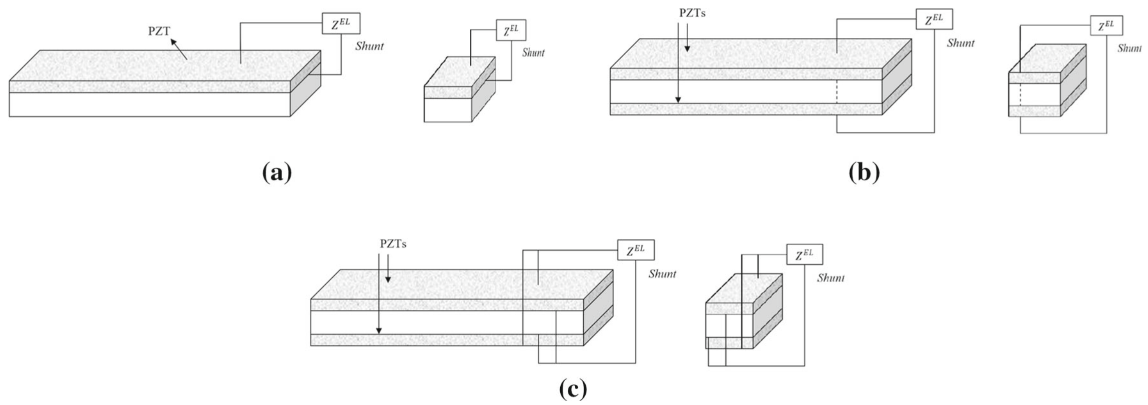



**Fig. 5** Structures and unit cell: **a** unimorph beam-shunt (RHS) and periodic unit cell (LHS); **b** bimorph beam-shunt in series (RHS) and periodic unit cell (LHS); **c** bimorph beam-shunt in parallel (RHS) and periodic unit cell (LHS)

tric properties are assumed to be $d_{31} = -175e^{-12}$m/V, $\kappa = 0.9$, $h_{31} = 0.31$, and $C_p^T = 200nF$. The parameters are assumed based on the work of [40] for validation purpose. Scope of this paper is limited to the analysis of only resonant circuits because, unlike the R, L, and C circuits, their impedance can be easily designed to specific frequencies or a targeted vibration mode. The structural vibration attenuation properties depend on the resonant behavior of the shunts. The resonant shunt circuits in series and parallel configuration are of the types LC, RL, and RCL. Transfer matrix method estimates the dispersion diagram from the dynamic stiffness of a unit cell of the periodic assembly and it is used to demonstrate the attenuation properties of the unimorph and bimorph beams. The receptance response is obtained from the beam in its full length. By applying the Transfer matrix method and assuming the periodicity of the structure, we can use only a unit-cell to obtain the dispersion diagram.

### 4.1 Shunt circuit inductor–capacitor, LC

For the general impedances of the inductor–capacitor (LC) circuit in series and parallel, we use an inductor of $L = 0.148H$ and a capacitor of $C = 1KnF$. These impedances work as the dynamic damper and are used to promote the vibrational and wave propagation control of the beam.

The shunt circuits LC in series and parallel configurations are known for their feasibility of passing electrical current filter because inductance and capacitance values are the same for series and parallel configurations. However, the effects of each configuration generate general impedances with peaks of different amplitudes in different tuned frequencies. The influence justifies the difference in series and parallel LC shunt configurations that piezoelectric capacitance exerts on the capacitance of the circuit shunt and vice versa. However, by applying the general impedances of Fig. 6a and b in the unimorph beam, one has the receptance frequency response functions (FRF) and dispersion diagrams, as shown in Fig. 7.

All dispersion diagrams present the wave propagating components (positive axis) in solid lines and evanescent component (negative axis) in dashed lines. Figure 7a and b demonstrate the resonances splitting into two peaks due to the inductive component of the shunt circuit. The result of Fig. 7b confirms that the location of the overall impedance is the factor of the resonance peak splitting at the target frequency. However, the receptance responses do not present any damping effect because of the missing resistor in this circuit. Figure 7c and d present the dispersion diagram containing two dispersive and two non-dispersive wave modes (both with real and imaginary parts, where real is the propagating waves and imaginary the evanescent). The dispersive modes or flexural mode (orange line) changes exactly at the tuned frequency induced by the impedance. This change increases its amplitude for the case with the circuit shunt LC in series. The non-dispersive or longitudinal modes (purple line) in both configurations shows no change. Note that the black curves in Figure 7c and d represent the responses corresponding to a beam without control. It can be noticed that each wavenumber presents a considerable imaginary part at the local resonances, which means that the corresponding waves propagate. This behavior induces vibration attenuation of the dynamic response.

By applying the general impedances of Fig. 6a and b in the bimorph beams in series and parallel, one has the FRFs and the dispersion diagrams shown in Fig 8. The FRFs of the bimorph beams connected to LC circuit shunt in series (s) and parallel (p) are showing in Fig. 8a and b. In the circuit with series topology, dynamic behavior is influenced most by the inductance, while that in the parallel topology is affected by the capacitance. Series and parallel LC circuits have similar behavior, but a different tuning frequency occurs in each configuration.

Figure 8d and f presents the dispersion diagrams of the bimorph beam connected to the shunt LC circuit in series with series and parallel topologies, respectively. Figures 8d and f show the dispersion diagrams of the bimorph beam connected to the shunt LC circuit in parallel with series and parallel topologies. In both cases,

**Fig. 6** Shunt circuit LC impedance configured in: **a** series and **b** parallel

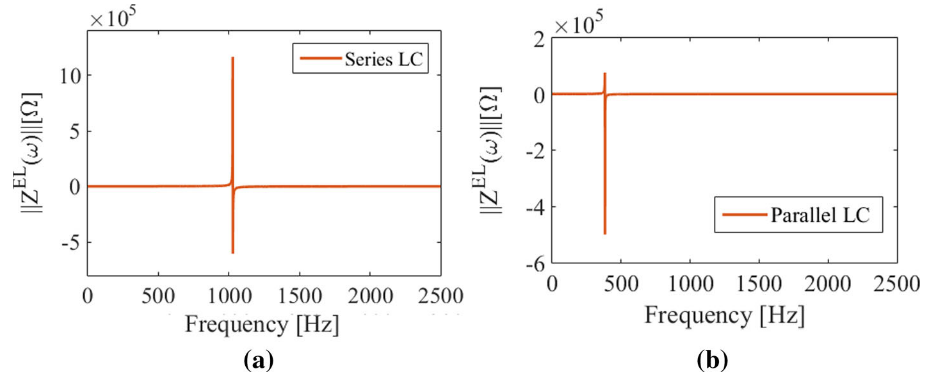
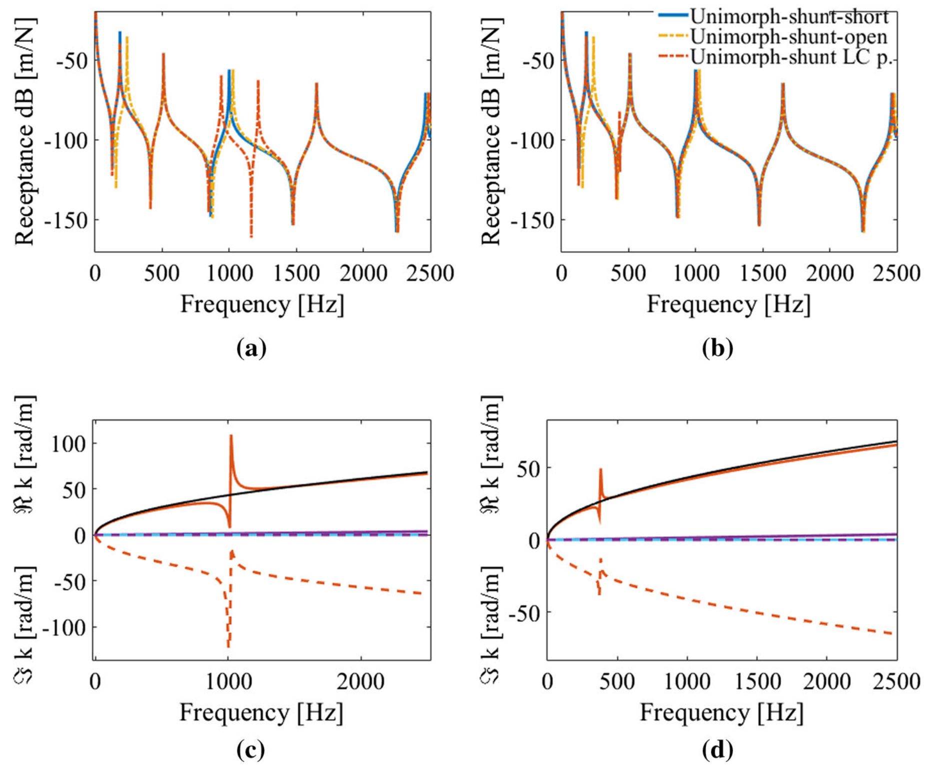



**Fig. 7** Unimorph beam connected to a shunt LC: **a** FRF for a shunt connected in series; **b** FRF for a shunt connected in parallel; **c** dispersion diagram with propagating component (positive axis) and evanescent component (negative axis) of the wavenumbers for a shunt connected in series; **d** dispersion diagram for a shunt connected in parallel




the dispersion diagrams contain two pairs of dispersive modes and one of non-dispersive nature. The wave shows a disruption in the propagation located at the tuning frequency in all dispersive and non-dispersive modes of the structure.

### 4.2 Shunt circuit resistor–inductor, RL

For the general impedances of the resistor–inductor (RL) circuit in series and in parallel employed in the beam vibration, and wave propagation attenuation and control, we use an inductor of $L = 0.148H$ and a resistor of $R = 33\Omega$. The impedance of the series configuration presents a wide peak at the target frequency. The parallel configuration impedance has a constant value over the frequency similar to a pure resistive shunt circuit [47]. However, by applying the general impedances of Fig. 9a and b to the unimorph beam, we can estimate the FRFs and the dispersion diagram shown in Fig. 10.

Figure 10a demonstrates the attenuation effect derived from the shunt RL circuit in the series configuration (due the damping effect imposed by the resistor). Although this attenuation effect does not occur in the shunt RL circuit in parallel, as shown in Fig. 10b, small damping in the first and third vibration modes is observed. The dispersion diagrams of the unimorph beam compared with conventional beam (black line) are demonstrated in Fig. 10c. The unimorph beam flexural wave mode (orange) presented an attenuation at the circuit-tuned frequency, while the longitudinal mode (purple line) shows no change. Aside from those two wave modes, a third one in the blue line shows propagation at the tuned frequency range. It is related to the impedance wave mode (blue line). The attenuation due to the shunt circuit appears in the FRFs (Fig. 9a)

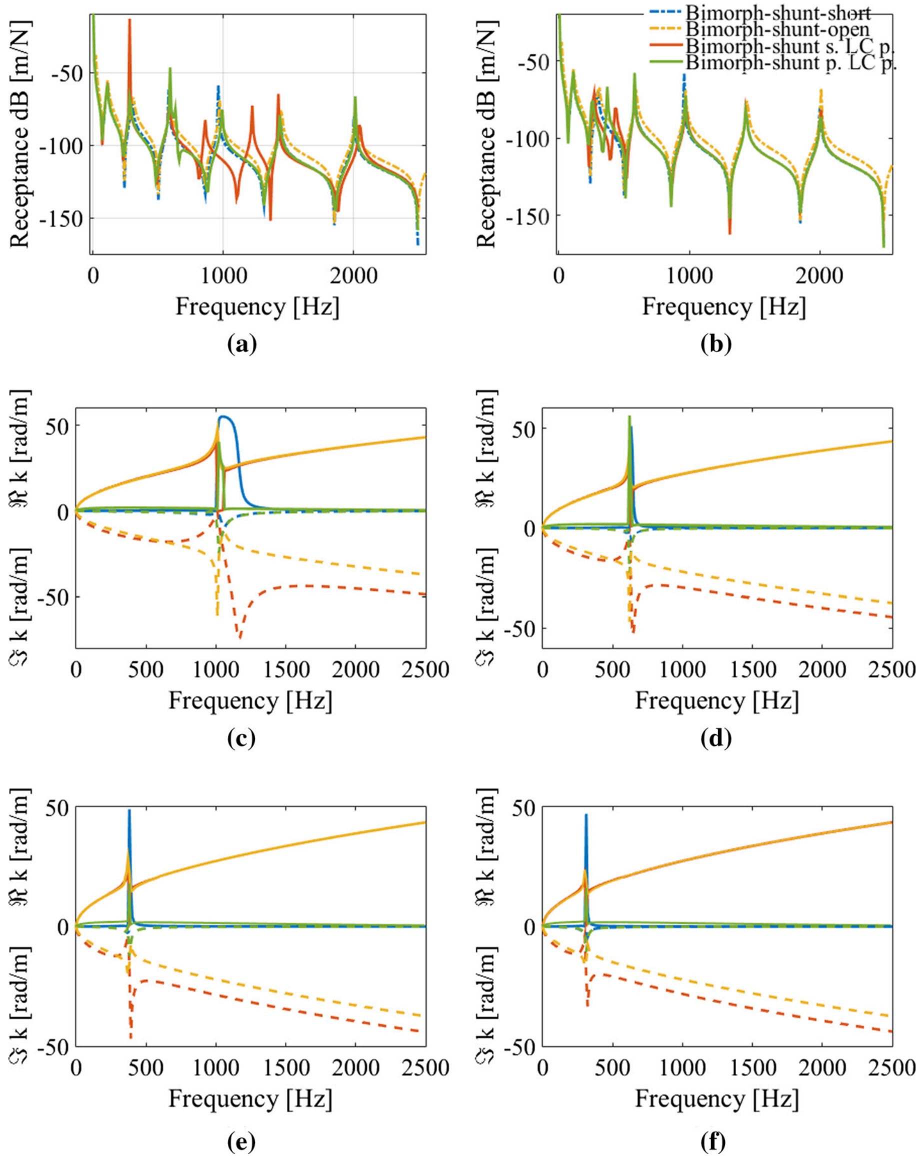



**Fig. 8** FRF and dispersion diagram of bimorph beam connected to a shunt LC: **a** FRF of shunt-series in series and parallel topology; **b** FRF of shunt-parallel in series and parallel topology; **c** Dispersion diagram of shunt-series in series; **d** dispersion diagram of shunt-series in parallel; **e** dispersion diagram of shunt-parallel in series; **f** dispersion diagram of shunt-parallel in parallel

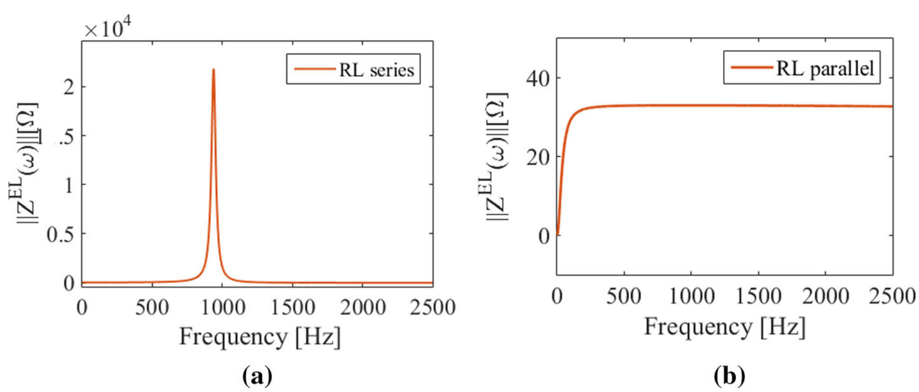



**Fig. 9** Shunt circuit RL impedance configured in: **a** series and **b** parallel

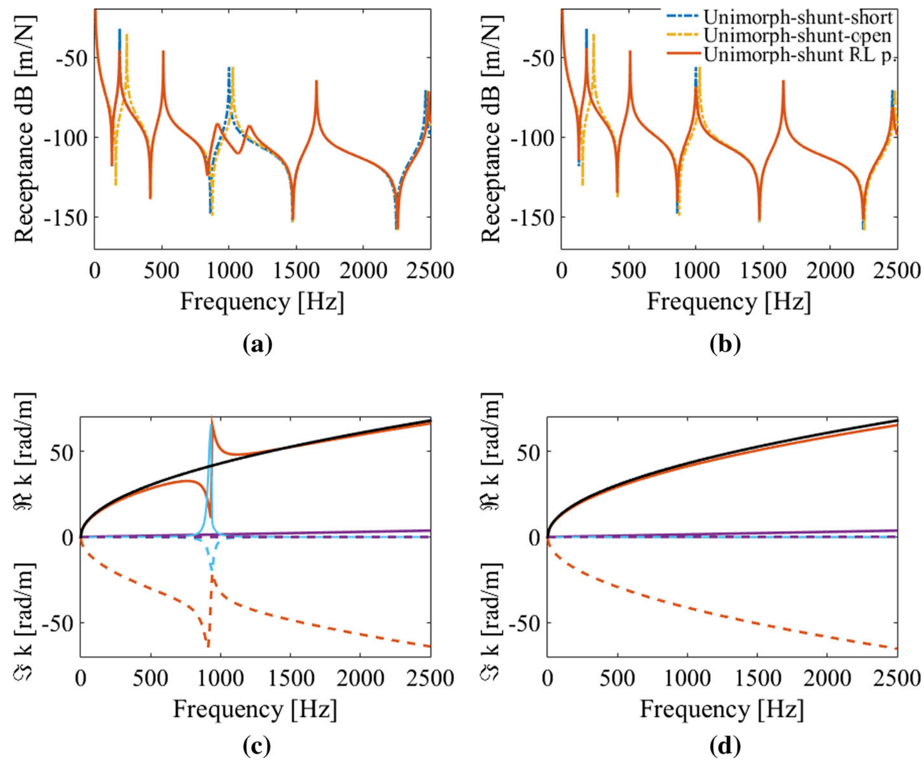



**Fig. 10** Unimorph beam connected to a shunt RL: **a** FRF for a shunt connected in series; **b** FRF for a shunt connected in parallel; **c** dispersion diagram with propagating component (positive axis) and evanescent component (negative axis) of the wavenumbers for a shunt connected in series; **d** dispersion diagram for a shunt connected in parallel

and dispersion diagram at the tuned frequency. The RL shunt in parallel has a constant impedance and it does not affect the unimorph beam dynamic responses, as shown in Fig. 10d.

By applying the impedances in series and parallel topology (Fig. 9a and b) onto the bimorph beam in series and a parallel configuration, one estimates the FRFs and the dispersion diagrams shown in Fig. 11. Figure 11a and b show the FRFs of the bimorph beam connected with the circuit shunt RL with series and parallel topology. Only the responses of the series shunt RL circuits present the vibration attenuation at the tuned frequency close to 1000 Hz. The parallel configuration remains unchangeable, as in the unimorph case. The dispersion diagrams present no change in all dispersive and non-dispersive modes for the cases with the shunt RL circuit in parallel, as showed in Fig. 11e and f. The impedance of the circuit shunt RL in series, shown in Fig. 11c and d, present attenuation in the flexural and longitudinal wave modes, specifically at shunt tuned frequency.

### 4.3 Shunt circuit resistor–inductor–capacitor, RLC

The impedance of the shunt resistor–inductor–capacitor (RLC) circuit is similar to the RL. Therefore, general impedances of the RLC circuit in series and parallel use values for inductor of $L = 0.148$ H, the resistor of $R = 33\Omega$, and capacitor of $C = 1$KnF. Figure 12 shows the shunt RLC impedances in series and parallel with resistors value of $33, 43$ and $63\Omega$. The resistors act as a damper in the circuit [42].

The circuits shunt RLC has a similar impedance behavior as the RL in series and configurations parallel, see Fig. 12. Also known as resonant shunts circuits, RL circuits become RLC circuits when coupled to a piezoelectric material. However, the difference is in the ability to adjust the stiffness. While RL is limited to the piezoelectric capacitance $C_p^T$, RLC is limited to its capacitor $C_{\mathrm{SH}}$, which can take an appropriate value to adjust the stiffness in the desired frequency range. Therefore, by applying the general resistor impedances $R = 33\Omega$ on the unimorph beam, we have the FRFs and the dispersion diagrams.

Figure 14 presents the FRFs of the bimorph beam connected to the shunt circuit in series and parallel, along with the dispersion diagrams. The attenuation effect caused by the series shunt RLC circuit shown in Fig. 13a happens similarly to the case provided by the series RL circuit. The wave propagation has similar behavior to the RL as presented in the dispersion diagram of the circuit shunt RLC in series and parallel (Fig. 13c and d) by applying the general impedances of Fig. 12a and b in the bimorph beam in series and parallel. In summary, the similarity of FRFs responses and dispersion diagrams between RL and RCL circuit is associated with the impedance homogeneity, influenced by the piezoelectric capacitance ($C_p^T$) and external capacitance $C = 1$KnF. However, diversification between the impedance of the shunt RL and RLC circuits can be provided with the presence of the specific capacitive component since this element gives adjustments for the location of peaks in certain frequency ranges.

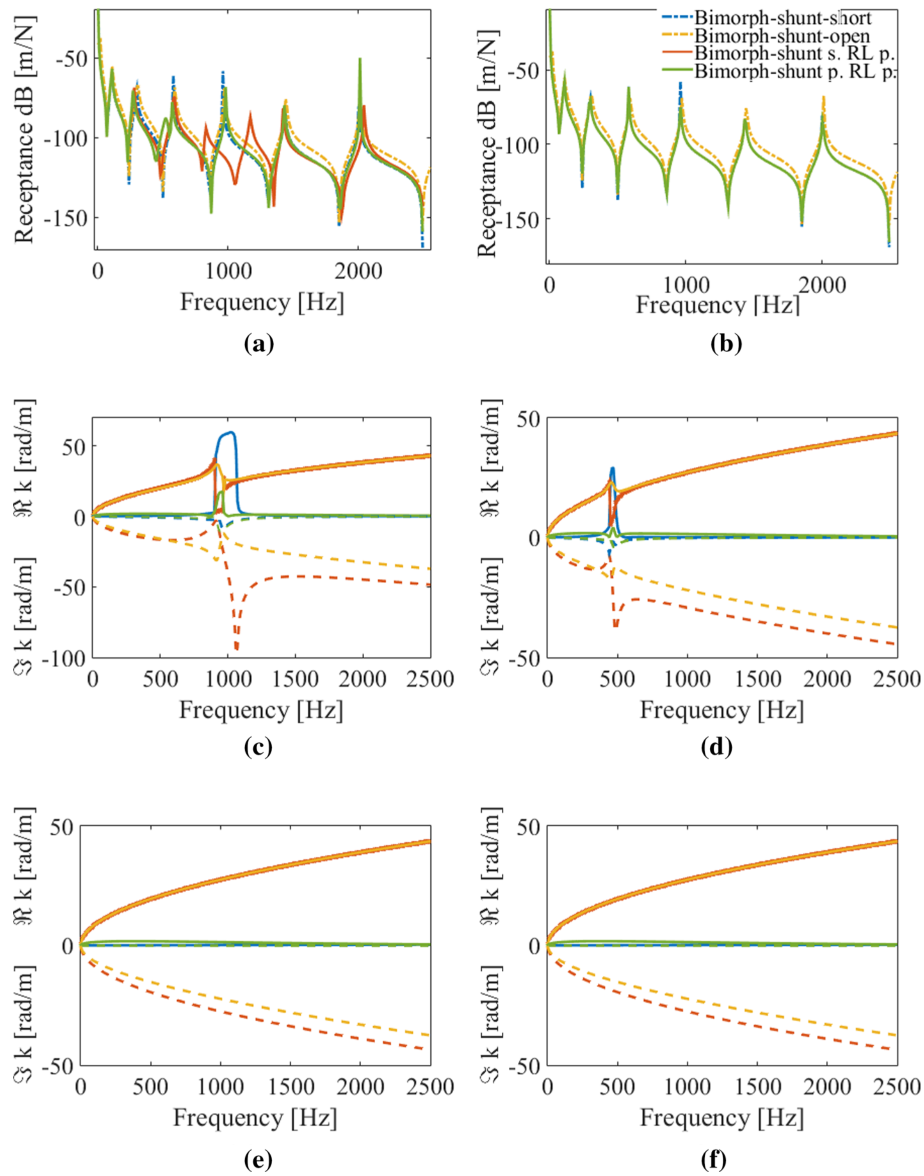



**Fig. 11** FRF and dispersion diagram of bimorph beam connected to a shunt RL: **a** FRF of shunt-series in series and parallel topology; **b** FRF of shunt-parallel in series and parallel topology; **c** dispersion diagram of shunt-series in series; **d** Dispersion diagram of shunt-series in parallel; **e** dispersion diagram of shunt-parallel in series; **f** dispersion diagram of shunt-parallel in parallel

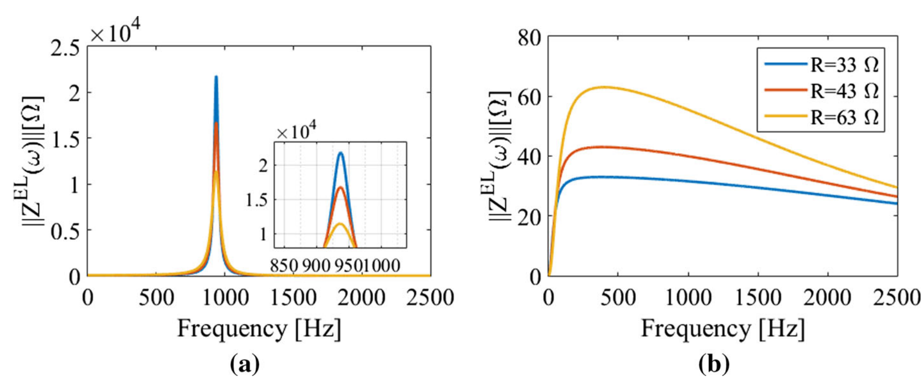



**Fig. 12** Shunt circuit RLC impedance configured in: **a** series and **b** parallel

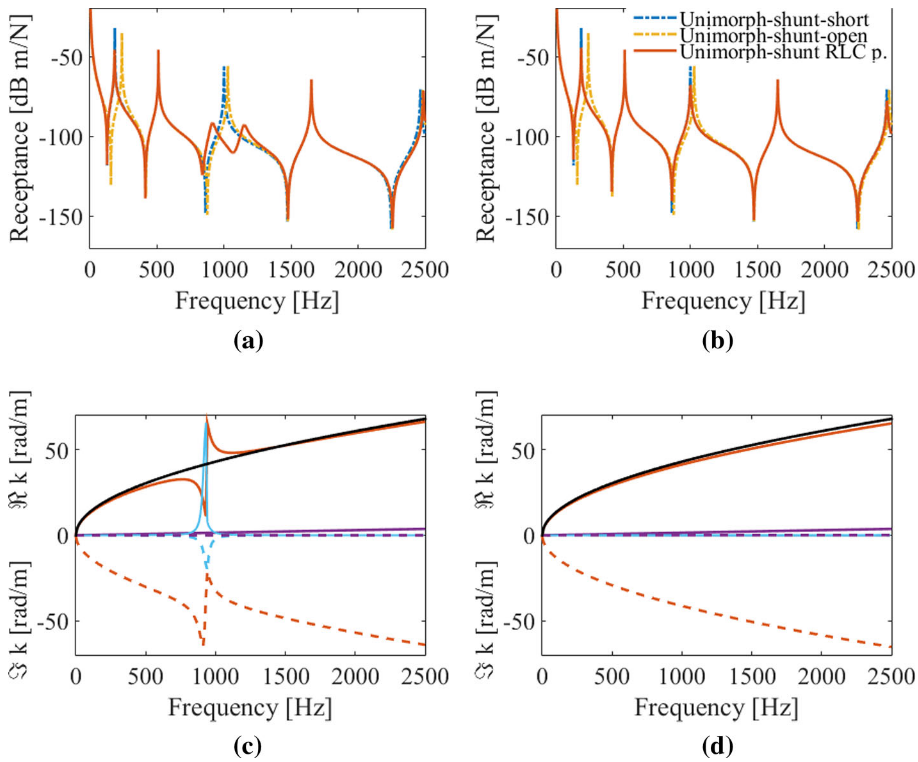



**Fig. 13** Unimorph beam connected to a shunt RLC: **a** FRF for a shunt connected in series; **b** FRF for a shunt connected in parallel; **c** dispersion diagram with propagating component (positive axis) and evanescent component (negative axis) of the wavenumbers for a shunt connected in series; **d** dispersion diagram for a shunt connected in parallel

## 5 Conclusions

This paper presents a dynamic analysis of unimorph and bimorph periodic piezo-beams with resonant RL, LC, and RCL shunt configurations. The main objective is to illustrate how the resonant shunt in different configurations can be used for wave attenuation and subsequent vibration reduction. For this purpose, a generic analytical formulation concerning efficient spectral piezo-beam element connected to the resonant shunt circuit is presented. One of the major advantages of the SEM is the small number of elements needed to represent a system, leading to a significant level of computational efficiency in dynamic and wave propagation analyses where a fine discretization is normally required following the conventional finite element-based approaches. The effects of the smart beams connected to the shunts circuit are demonstrated by the FRF amplitudes attenuated around the tuning frequencies. The effective flexural wavenumber presented by the dispersion diagrams for the smart beams produces a typical locally resonant behavior at the same attenuation frequencies as the FRFs, indicating a direct effect of the resonating circuits and the vibration attenuation. Based on the comprehensive demonstration presented in this article, the spectral element approach shows adequate potential for further analysis and design of more complex configurations of smart structures (/ metamaterials) in an accurate, yet efficient framework.

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## A Transfer matrix method

Transfer matrix method allows us to exploit the dynamic stiffness matrix of a single piezo-beam for analyzing periodic arrangements of such beams. The spectral transfer matrix method [32] is an analytical spectral approach of the system in the state-space form to directly obtain the transfer matrix of the structure. From dynamic system relationship, a frequency-domain state-vector equation is obtained as

$$\frac{d\hat{\mathbf{y}}(x)}{\mathrm{d}x} = \mathbf{A}\left(\omega\right)\hat{\mathbf{y}}(x) \qquad \left(0 \leq x \leq L\right), \tag{40}$$

where $\mathbf{A}(\omega)$ is the system coefficient matrix with $[2n \times 2n]$ degrees of freedom, and $\hat{\mathbf{y}}(x) = \{\hat{\mathbf{d}} \quad \hat{\mathbf{F}}\}^{\mathbf{T}}$ is the state vector with $\hat{\mathbf{d}}$ as the spectral nodal displacements and $\hat{\mathbf{F}}$ the spectral nodal forces vectors. The general solution of Eq. (40) for a structural element of length $L$ is given by

$$\hat{\mathbf{y}}\left(L\right) = \mathbf{e}^{\mathbf{A}L}\hat{\mathbf{y}}\left(\mathbf{0}\right) = \mathbf{T}\left(\omega\right)\hat{\mathbf{y}}\left(\mathbf{0}\right), \tag{41}$$

where $\mathbf{T}(\omega) = \mathbf{e}^{\mathbf{A}L}$ is the transfer matrix, in its exponential form. If all eigenvectors of the matrix $\mathbf{A}$ are linearly independent, the transfer matrix can be written as

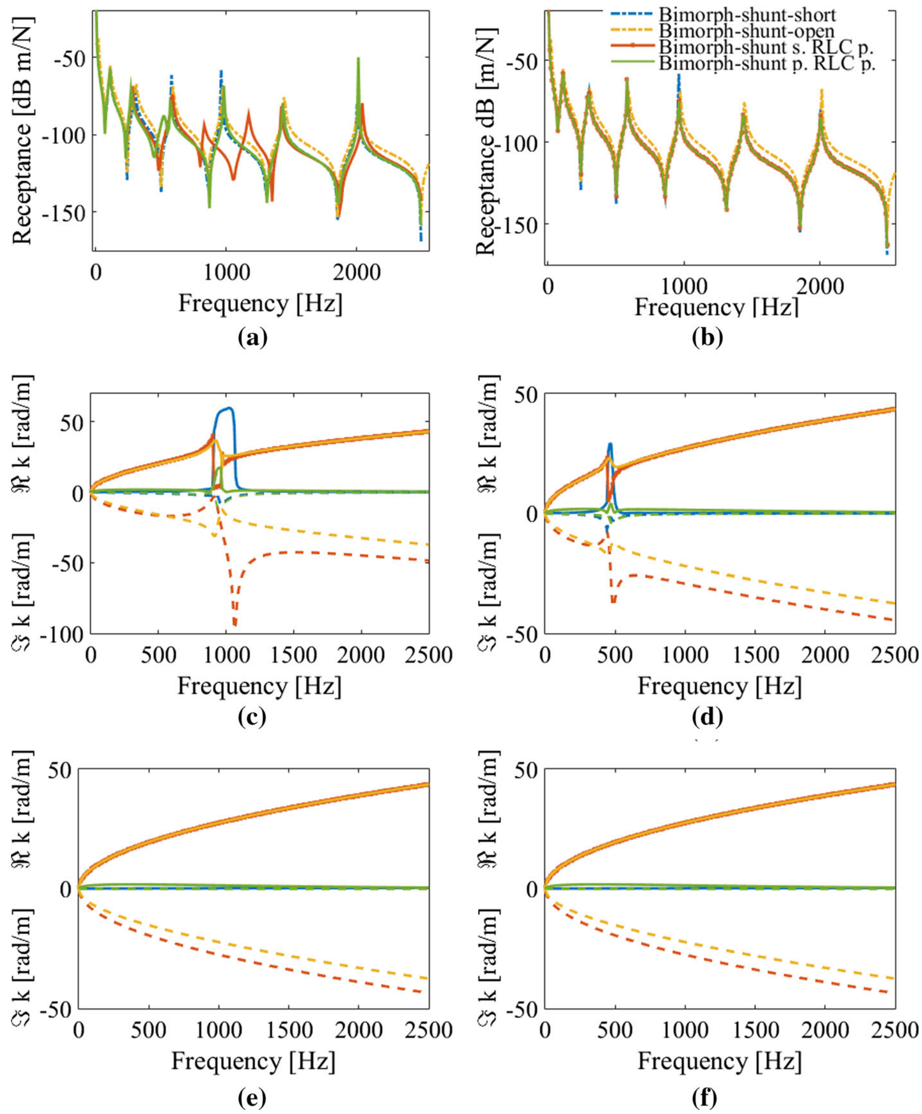


**Fig. 14** FRF and dispersion diagram of bimorph beam connected to a shunt RLC: **a** FRF of shunt-series in series and parallel topology; **b** FRF of shunt-parallel in series and parallel topology; **c** dispersion diagram of shunt-series in series; **d** dispersion diagram of shunt-series in parallel; **e** dispersion diagram of shunt-parallel in series; **f** dispersion diagram of shunt-parallel in parallel

$$\begin{Bmatrix} u_r \\ -F_r \end{Bmatrix} = \begin{bmatrix} -S_{lr}^{-1}S_{ll} & -S_{lr}^{-1} \\ S_{rl} - S_{rr}S_{lr}^{-1}S_{ll} & -S_{rr}S_{lr}^{-1} \end{bmatrix} \begin{Bmatrix} u_l \\ F_l \end{Bmatrix}$$
$$or \qquad \mathbf{q}_r = \mathbf{T}(\omega)\mathbf{q}_l, \tag{42}$$

where $\mathbf{T}(\omega)$ relates the left $(l)$ state vector $\mathbf{q}_l$ with the right $(r)$ state vector $\mathbf{q}_r$ of the unit-cell [46]

$$\mathbf{T}\begin{Bmatrix} u_l \\ F_l \end{Bmatrix} = e^{\mu} \begin{Bmatrix} u_r \\ F_r \end{Bmatrix} \qquad or \quad \mathbf{Tq} = e^{\mu}\mathbf{q}, \tag{43}$$

$\mathbf{A} = e^{\mu}$ is the corresponding eigenvector matrix and $\mathbf{q}$ are the corresponding eigenvectors derived from the Bloch wave eigenproblem with $\mu$ being the eigenvalues.

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